

LUMMAC2 THREAT HUNTING LAB

Hypothesis-Driven Threat Hunting Using Microsoft Sentinel & Sysmon

IOC Simulation, Endpoint Telemetry Generation & KQL-Based Threat Hunt

Lab Type: Hypothesis-Driven Threat Hunting (CISA Advisory AA25-141b)

CISA Advisory: Threat Actors Deploy LummaC2 Malware to Exfiltrate Sensitive Data

Platform: Microsoft Sentinel + Sysmon + Microsoft Defender XDR

VM / Host: Web01 -- Windows Server 2022 (Public IP: 20.5.48.127)

Simulation File: LummaC2.exe (safe -- launches calc.exe only, no malicious code)

Telemetry Source: Sysmon v15.21 (Event ID 1, 3, 7, 11, 13, 22) + Windows Security Events

DCR Created: Sysmon-DCR -- Custom Windows Logs: Microsoft-Windows-Sysmon/Operational!*

IOC Simulated: 94.158.244.69 (nslookup) | LummaC2.exe process name | iphlapi.dll / winhttp.dll

Hunting Queries: 5 KQL hunts mapped to MITRE ATT&CK and CISA advisory indicators

Hypothesis: What if LummaC2 was executed in this environment -- how would I detect it?

Outcome: Simulated telemetry confirmed in Sentinel. Hunt 1 returned 3 results. Advisory hunts returned no results (expected -- simulation only).

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1. Lab Overview

This lab demonstrates hypothesis-driven threat hunting using Microsoft Sentinel and Sysmon, aligned with the CISA cybersecurity advisory AA25-141b: "Threat Actors Deploy LummaC2 Malware to Exfiltrate Sensitive Data from Organizations."

The lab follows a structured threat hunting methodology: starting from a hypothesis that LummaC2-like activity may have occurred in the environment, then creating a safe simulation executable (LummaC2.exe) to generate endpoint telemetry, installing Sysmon for deep endpoint visibility, ingesting Sysmon logs into Sentinel via a custom DCR, and writing 5 KQL hunting queries mapped to real CISA advisory indicators including known LummaC2 DLLs, domains, IOC IP addresses, and process execution patterns.

Important: LummaC2.exe contains NO malicious functionality. It is a PowerShell script compiled to an executable that only launches calc.exe. It was intentionally named LummaC2.exe to generate a named indicator of compromise (IOC) in the environment for hunting and detection validation purposes only.

2. Lab Objectives

- * Simulate a safe LummaC2-style indicator by creating and executing a named executable (LummaC2.exe).
- * Install Sysmon v15.21 on Web01 to provide deep endpoint telemetry (process creation, network, DNS, DLL loads).
- * Create a custom Data Collection Rule (DCR) to ingest Sysmon logs into Microsoft Sentinel.
- * Verify Sysmon log ingestion in Sentinel Logs using KQL.
- * Simulate IOC investigation by running nslookup against a known LummaC2 IP (94.158.244.69).
- * Write 5 KQL hunting queries based on CISA Advisory AA25-141b indicators.
- * Demonstrate the full hypothesis-driven threat hunting workflow: hypothesis -> simulation -> telemetry -> hunt -> findings.

3. Threat Context -- CISA Advisory AA25-141b

This lab is based on the CISA advisory: "Threat Actors Deploy LummaC2 Malware to Exfiltrate Sensitive Data from Organizations" (<https://www.cisa.gov/news-events/cybersecurity-advisories/aa25-141b>).

What is LummaC2?

LummaC2 (also known as Lumma Stealer) is an information-stealing malware sold as Malware-as-a-Service (MaaS), first observed in August 2022. It is developed in C and targets cryptocurrency wallets, browser credentials, login data, and other sensitive information. It regularly updates to evade detection and expand its capabilities.

Also Known As	Lumma Stealer LummaC2 Stealer
Type	Information Stealer (MaaS)
Origin	ex-USSR

First Seen	August 1, 2022
Last Seen	June 22, 2026
Primary Targets	Cryptocurrency wallets, browser credentials, login data, sensitive files
Delivery Method	Phishing, masquerading, ingress tool transfer
C2 Method	Web-based command and control
Notable DLLs	iphlpapi.dll (IP Helper API), winhttp.dll (Windows HTTP Services)
IOC IP (lab)	94.158.244.69

MITRE ATT&CK Techniques (per CISA Advisory)

Technique ID	Technique Name	Tactic
T1566	Phishing	Initial Access
T1036	Masquerading	Defense Evasion
T1217	Browser Information Discovery	Discovery
T1119	Automated Collection	Collection
T1102	Web Service (C2)	Command & Control
T1105	Ingress Tool Transfer	Command & Control
T1041	Exfiltration Over C2 Channel	Exfiltration

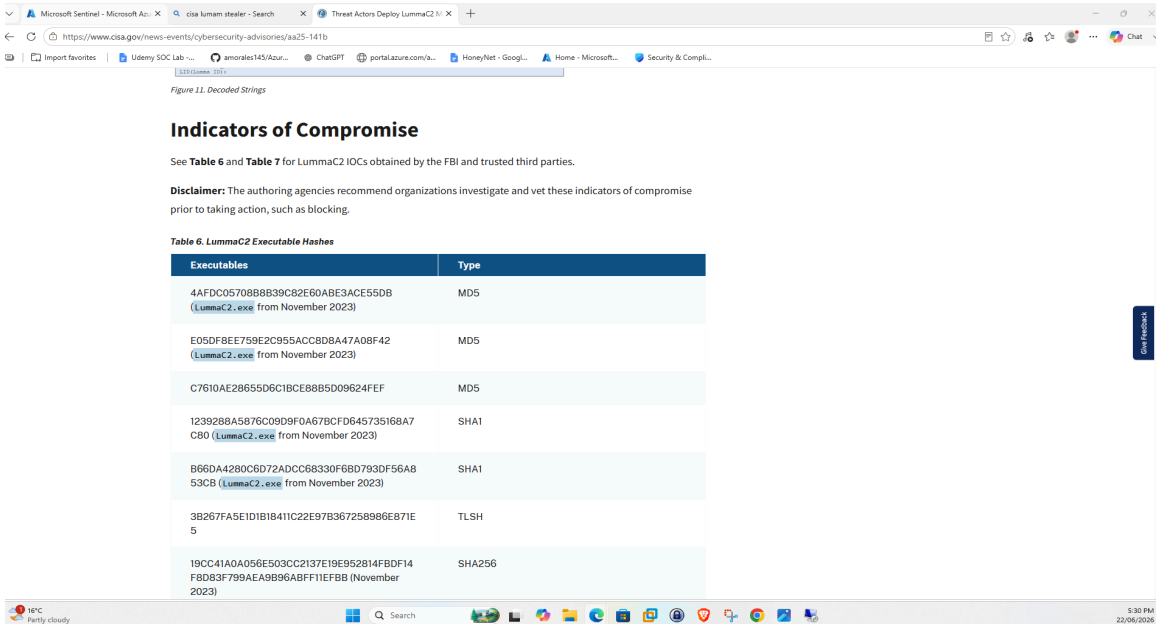


Fig 1 -- CISA Advisory AA25-141b: LummaC2 Indicators of Compromise

4. Threat Hunting Hypothesis

Hypothesis

"LummaC2-like activity may have executed in the environment. The goal of this hunt is to identify suspicious execution, DLL loading, network activity, DNS lookups, and possible download or transfer behavior related to LummaC2 tradecraft."

This hypothesis is based on the CISA advisory AA25-141b and drives the hunting queries, simulation design, and telemetry collection strategy used throughout this lab.

5. Phase 1 -- LummaC2 Simulation Setup

A safe simulation executable was created to generate endpoint telemetry under the LummaC2.exe process name. The executable contains no malicious code -- it only prints output and launches Calculator (calc.exe).

5.1 Step 1 -- Create PowerShell Simulation Script

File: LummaStealer_Simulation.ps1

```
Write-Output "Lumma Stealer Simulation Started"
Start-Process calc.exe
Write-Output "Simulation Complete"
```

5.2 Step 2 -- Install PS2EXE Module

PS2EXE converts the PowerShell script into a Windows executable (.exe) so it generates process telemetry under a realistic filename.

```
Install-Module PS2EXE -Scope CurrentUser
Import-Module PS2EXE
```

5.3 Step 3 -- Compile PowerShell Script to Executable

```
Invoke-PS2EXE `
  -InputFile .\LummaStealer_Simulation.ps1 `
  -OutputFile .\LummaC2.exe
```

Output confirmed: "Output File C:\Users\bikashsoc1\Desktop\LummaC2.exe written"

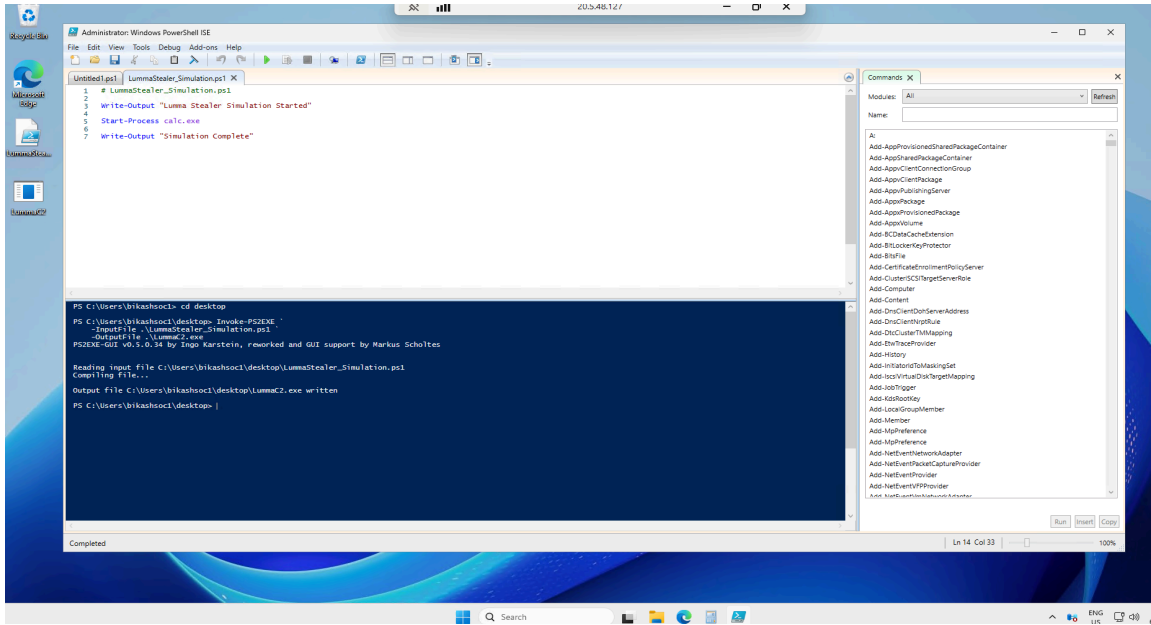


Fig 2 -- PowerShell ISE: LummaStealer_Simulation.ps1 open + PS2EXE compile complete -- LummaC2.exe written to Desktop

5.4 Step 4 -- Execute LummaC2.exe

```
. \LummaC2.exe
```

LummaC2.exe executed successfully. calc.exe launched as a child process. Process creation telemetry generated by Microsoft Defender and Sysmon.

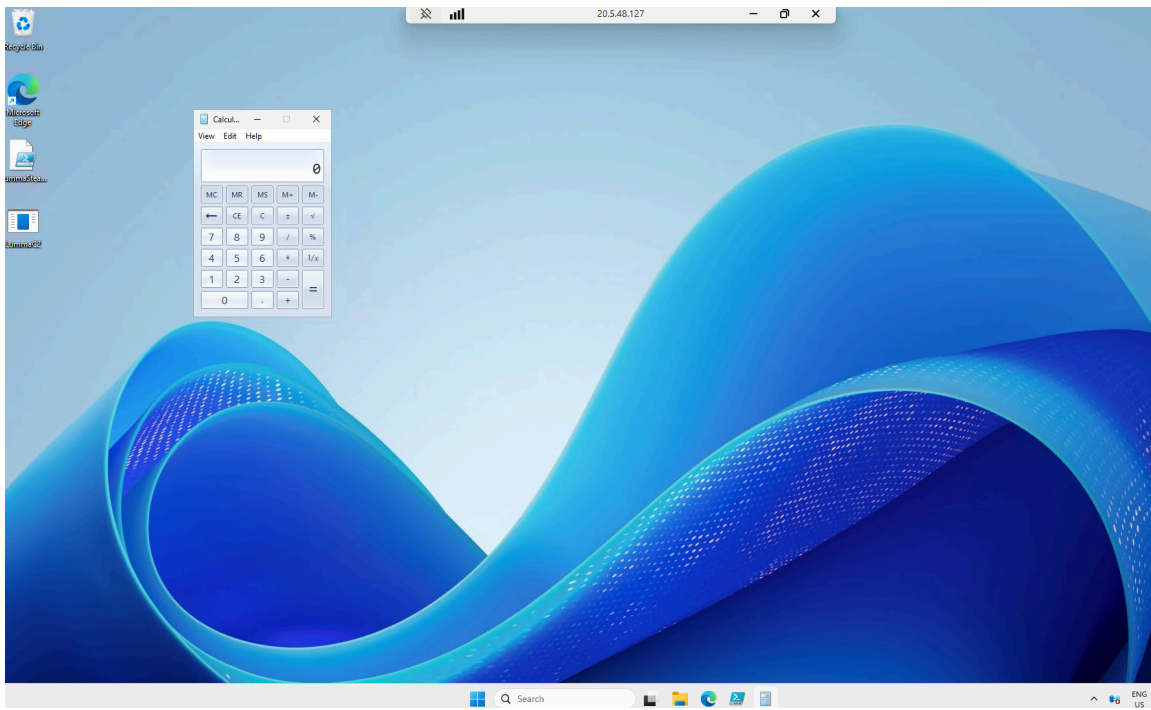


Fig 3 -- Web01 Desktop: LummaC2.exe executed -- calc.exe opened as child process (Windows 11 desktop visible, LummaC2.exe icon on desktop)

6. Phase 2 -- Sysmon Installation & Configuration

Sysmon (System Monitor) was installed on Web01 to provide deep endpoint telemetry beyond what standard Windows Event logging captures. Sysmon tracks process creation, network connections, DNS queries, DLL image loads, file creation, and registry modifications.

6.1 Download Sysmon

Sysmon v15.21 was downloaded from the official Microsoft Sysinternals page:

```
https://learn.microsoft.com/sysinternals/downloads/sysmon
```

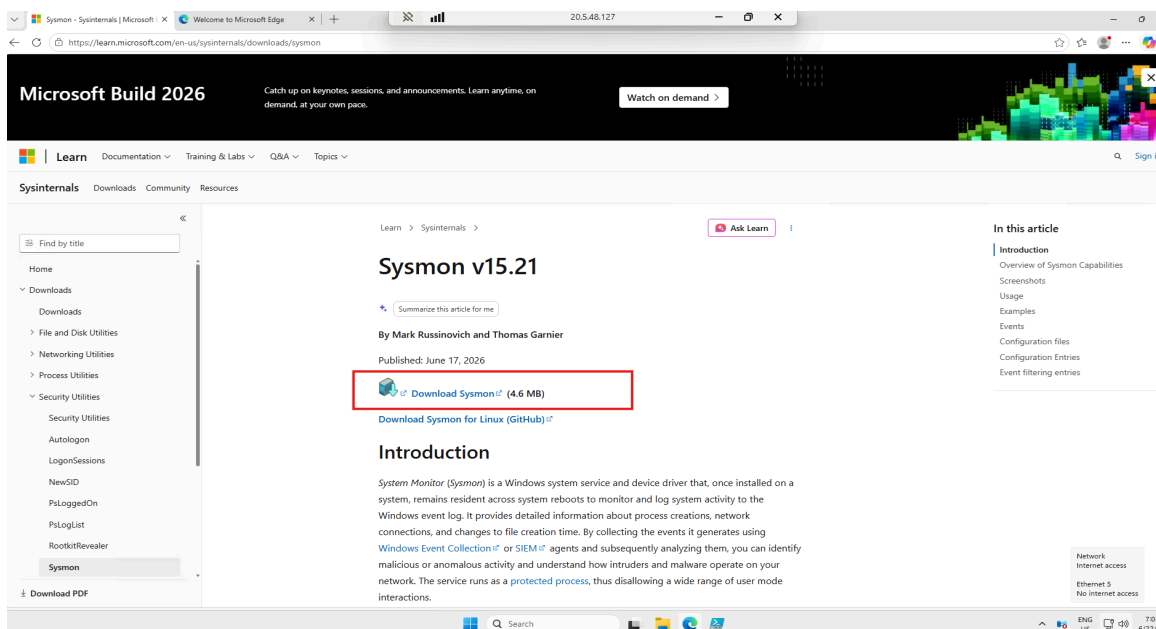


Fig 4 -- Microsoft Learn Sysinternals: Sysmon v15.21 download page -- Download Sysmon (4.6 MB) highlighted

6.2 Install Sysmon

```
cd C:\Users\bikashsoc1\Downloads\Sysmon
.\Sysmon64.exe -accepteula -i
```

Installation output confirmed:

- * Sysmon installed.
- * SysmonDrv installed.
- * Starting SysmonDrv.
- * SysmonDrv started.
- * Starting Sysmon64.
- * Sysmon64 started.

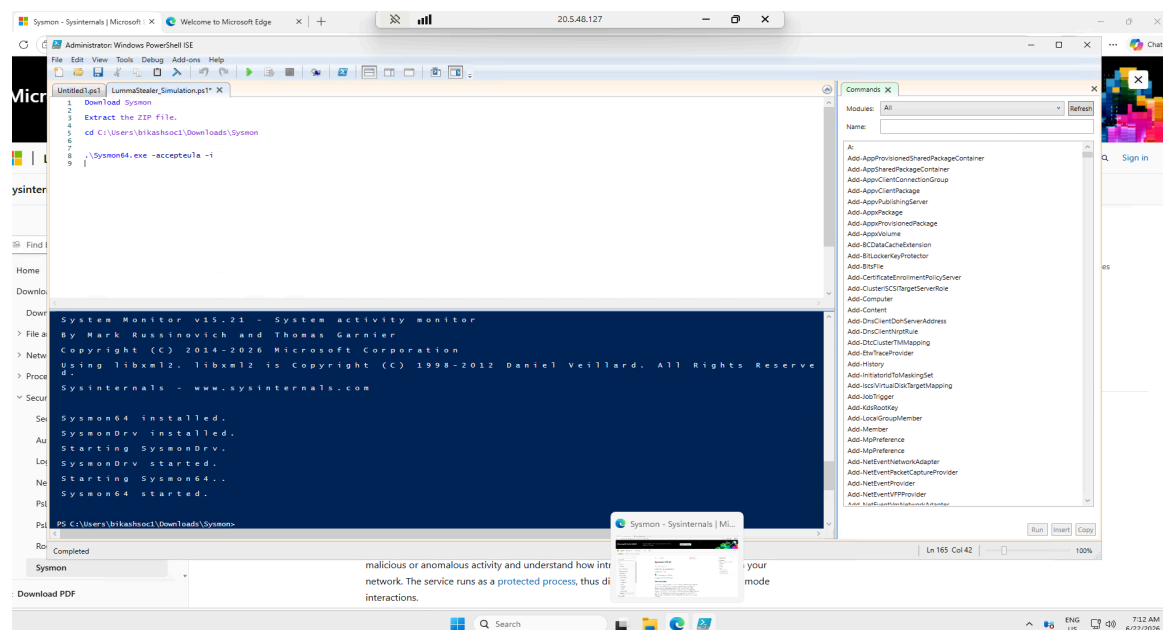


Fig 5 -- PowerShell: Sysmon64.exe -accepteula -i -- Installation output confirming Sysmon installed, SysmonDrv started, Sysmon64 started

6.3 Verify Sysmon Service

```
Get-Service Sysmon64
```

Status	Running
Name	Sysmon64
DisplayName	Sysmon64

6.4 Verify Sysmon Logs in Event Viewer

Navigation: Event Viewer > Applications and Services Logs > Microsoft > Windows > Sysmon > Operational

Sysmon Event ID	Description
Event ID 1	Process Creation -- captures process name, path, hash, parent process, command line
Event ID 3	Network Connection -- captures source/dest IP, port, process name
Event ID 7	Image Loaded -- captures DLL loads with filename, hash, signing details
Event ID 11	File Creation -- captures newly created files with path and hash
Event ID 13	Registry Modification -- captures registry value changes
Event ID 22	DNS Query -- captures DNS queries with process name and queried domain

7. Phase 3 -- Generate Telemetry & Ingest into Sentinel

7.1 Generate Telemetry

LummaC2.exe was re-executed to generate Sysmon telemetry after Sysmon was installed:

```
.\LummaC2.exe
```

Sysmon Event ID 1 (Process Creation) was confirmed in Event Viewer under Sysmon/Operational, showing:

- * Image: C:\Users\bikashsoc1\Desktop\LummaC2.exe
- * ProcessId: 4138
- * Log Name: Microsoft-Windows-Sysmon/Operational
- * Task Category: Process Create (rule: ProcessCreate)
- * Computer: Web01

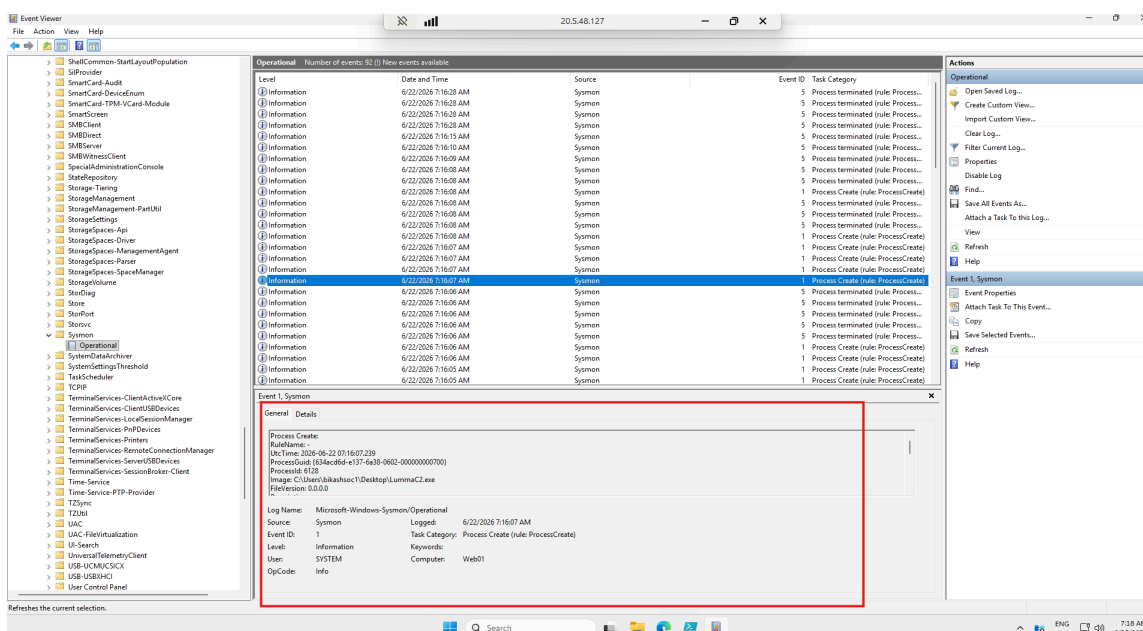


Fig 6 -- Event Viewer: Sysmon/Operational -- Event ID 1 Process Create highlighted for LummaC2.exe -- ProcessId 4138, Image path confirmed, Computer: Web01

7.2 Create Sysmon DCR -- Ingest Logs into Sentinel

A new Data Collection Rule was created to collect Sysmon logs from Web01 and send them to the Bik-SOC-Lab Log Analytics Workspace / Microsoft Sentinel.

DCR Name	Sysmon-DCR
Resource Group	Security-BikSecOps
Log Source	Custom Windows Logs: Microsoft-Windows-Sysmon/Operational!*
Target Workspace	Bik-SOC-Lab
Status	Deployed successfully

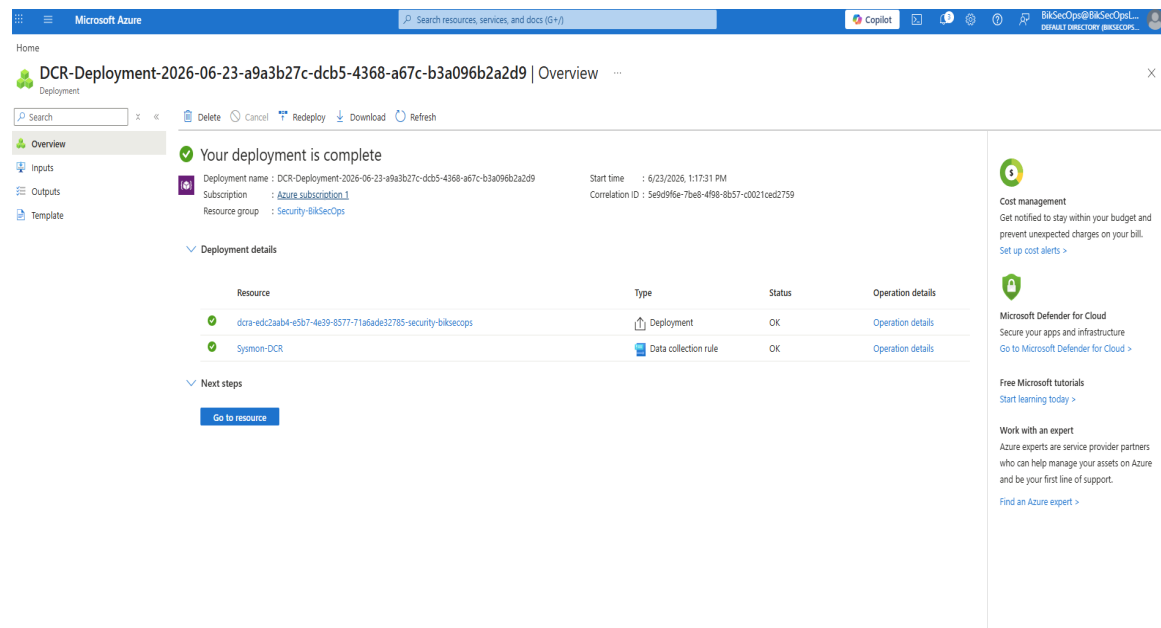


Fig 7 -- Azure: Sysmon-DCR Data Collection Rule deployment complete -- DCR Deployment OK, Sysmon-DCR Data collection rule OK

7.3 Verify Sysmon Logs in Sentinel

KQL was used to verify that Sysmon logs were flowing into Sentinel:

```
Event
| where EventLog == "Microsoft-Windows-Sysmon/Operational"
| where EventID == 1
| take 20
```

Results confirmed 13+ rows from Microsoft-Windows-Sysmon/Operational, Source: Microsoft-Windows-Sysmon, Computer: Web01, EventCategory: 1.

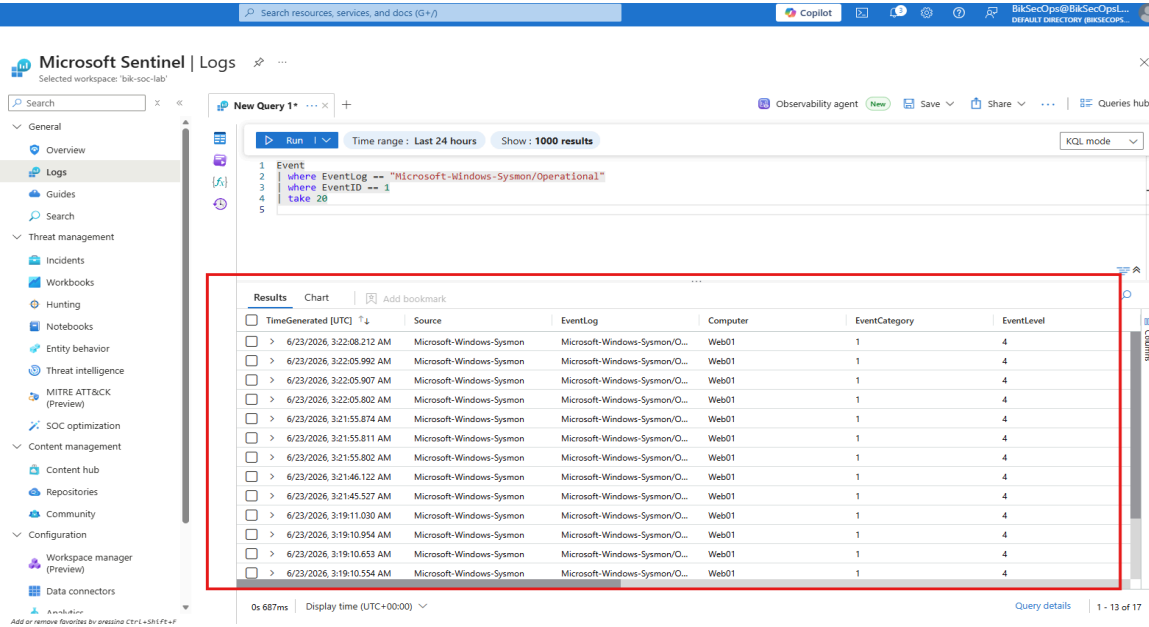


Fig 8 -- Sentinel Logs: Sysmon EventID 1 logs confirmed -- 13 results from Microsoft-Windows-Sysmon/Operational, Computer: Web01

8. Phase 4 -- IOC Simulation: nslookup LummaC2 IP

To simulate LummaC2 network activity, nslookup was performed against a known LummaC2-associated IP address (94.158.244.69) from Web01. This generates a Sysmon Event ID 1 (process creation for nslookup.exe) which can be hunted in Sentinel.

IOC IP Address	94.158.244.69
Resolved Hostname	no-rdns.mivocloud.com
Resolved Address	168.63.129.16 (Azure DNS)
Source	CISA Advisory AA25-141b / any.run LummaC2 analysis
Sysmon Event Generated	Event ID 1 -- nslookup.exe process creation with 94.158.244.69 in command line
Command Run	C:\Users\bikashsoc1> nslookup 94.158.244.69

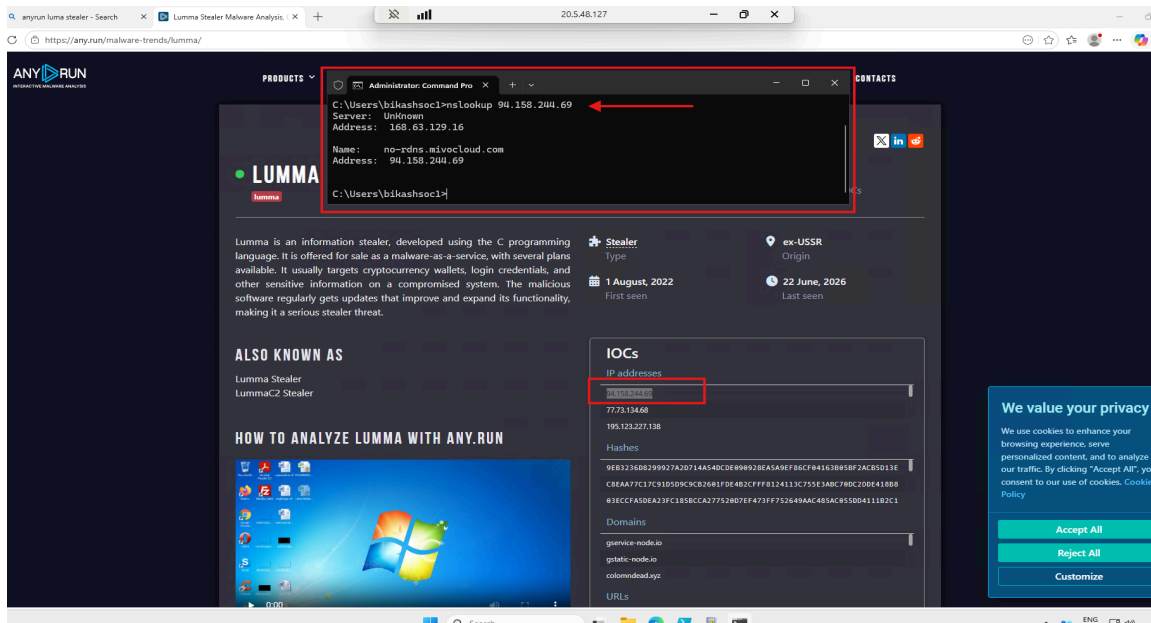


Fig 9 -- Web01 CMD: nslookup 94.158.244.69 -- resolves to no-rdns.mivocloud.com | any.run showing Lumma Stealer malware profile with IOC IP 94.158.244.69 highlighted

9. Phase 5 -- KQL Threat Hunting Queries

Five hunting queries were written based on the CISA Advisory AA25-141b indicators, following the hypothesis-driven hunting methodology. Queries target: process execution, DLL image loads, network connections, DNS queries, and suspicious download activity.

Hunt 1 -- LummaC2 Process Execution (Simulated IOC)

Hunts for Sysmon Event ID 1 (Process Creation) where the process name matches LummaC2.exe -- the simulated IOC filename.

```
Event
| where TimeGenerated > ago(24h)
| where EventLog == "Microsoft-Windows-Sysmon/Operational"
| where EventID == 1
| where RenderedDescription has_any ("LummaC2.exe")
| project
    TimeGenerated,
    Computer,
    SysmonEventID = EventID,
    Activity = "LummaC2 Simulation Process Execution",
    RenderedDescription
| order by TimeGenerated desc
```

Result: 3 rows returned -- confirmed LummaC2.exe process creation events on Web01 with full RenderedDescription showing Image path, ProcessId, ParentImage, CommandLine, and hashes.

The screenshot displays the Microsoft Sentinel interface with a KQL query executed. The query filters for events where the EventLog is 'Microsoft-Windows-Sysmon/Operational', EventID is 1, and the RenderedDescription contains 'LummaC2.exe'. The results table shows three rows of data, each representing a process creation event for LummaC2 Simulation Process Execution on the Web01 computer. The first row is expanded, showing detailed telemetry including the Process Create RuleName, UtcTime, ProcessGuid, ProcessId, Image, FileVersion, WinBuild, Description, and the full command line for the process.

TimeGenerated [UTC]	Computer	SysmonEventID	Activity	RenderedDescription
6/23/2026, 3:26:41.274 AM	Web01	1	LummaC2 Simulation Process Execution	Process Create: RuleName: - UtcTime: 2026-06-23 03:26:41.249 ProcessGuid: {634acd6d-fc1f-6a39-c901-000000000000} ProcessId: 6776 Image: C:\Windows\System32\calc.exe FileVersion: 10.0.26100.1882 (WinBuild.160101.0800) Description: Window...
6/23/2026, 3:26:40.312 AM	Web01	1	LummaC2 Simulation Process Execution	Process Create: RuleName: - UtcTime: 2026-06-23 03:26:40.308 ProcessGuid: {634acd6d-fc1f-6a39-c701-000000000000} ProcessId: 9060 Image: C:\Windows\System32\cmd.exe FileVersion: 10.0.26100.1882 (WinBuild.160101.0800) Description: Command...
6/23/2026, 3:26:40.290 AM	Web01	1	LummaC2 Simulation Process Execution	Process Create: RuleName: - UtcTime: 2026-06-23 03:26:40.285 ProcessGuid: {634acd6d-fc1f-6a39-c601-000000000000} ProcessId: 4380 Image: C:\Users\Bikashoc1\Desktop\LummaC2.exe FileVersion: 1.0.0.0 Description: LummaC2 Simulation Process Execution

Fig 10 -- Sentinel Logs: Hunt 1 KQL results -- 3 rows, LummaC2 Simulation Process Execution activity, RenderedDescription showing LummaC2.exe execution details on Web01

The screenshot displays the Microsoft Sentinel interface with the same KQL query as Figure 10. The results table shows three rows of data, each representing a process creation event for LummaC2 Simulation Process Execution on the Web01 computer. The first row is expanded, showing detailed telemetry including the Process Create RuleName, UtcTime, ProcessGuid, ProcessId, Image, FileVersion, WinBuild, Description, and the full command line for the process.

TimeGenerated [UTC]	Computer	EventID	RenderedDescription
6/23/2026, 3:26:41.274 AM	Web01	1	Process Create: RuleName: - UtcTime: 2026-06-23 03:26:41.249 ProcessGuid: {634acd6d-fc1f-6a39-c901-000000000000} ProcessId: 6776 Image: C:\Windows\System32\calc.exe FileVersion: 10.0.26100.1882 (WinBuild.160101.0800) Description: Window...
6/23/2026, 3:26:40.312 AM	Web01	1	Process Create: RuleName: - UtcTime: 2026-06-23 03:26:40.308 ProcessGuid: {634acd6d-fc1f-6a39-c701-000000000000} ProcessId: 9060 Image: C:\Windows\System32\cmd.exe FileVersion: 10.0.26100.1882 (WinBuild.160101.0800) Description: Command...
6/23/2026, 3:26:40.290 AM	Web01	1	Process Create: RuleName: - UtcTime: 2026-06-23 03:26:40.285 ProcessGuid: {634acd6d-fc1f-6a39-c601-000000000000} ProcessId: 4380 Image: C:\Users\Bikashoc1\Desktop\LummaC2.exe FileVersion: 1.0.0.0 Description: LummaC2 Simulation Process Execution

Fig 11 -- Sentinel Logs: Hunt 1 expanded results -- Process Create entries for LummaC2.exe showing full process telemetry (PID, hashes, parent process, command line)

IOC Investigation Hunt -- nslookup 94.158.244.69

Hunts for any Sysmon event referencing the LummaC2-associated IP address in the RenderedDescription, catching the nslookup simulation.

```

Event
| where TimeGenerated > ago(24h)
| where EventLog == "Microsoft-Windows-Sysmon/Operational"
| where RenderedDescription contains "94.158.244.69"
| project TimeGenerated, Computer, EventID, RenderedDescription
| order by TimeGenerated desc

```

Result: 1 row returned -- Sysmon Event ID 1 for nslookup.exe (C:\Windows\System32\nslookup.exe) with 94.158.244.69 in the RenderedDescription, confirming the IOC lookup was captured.

The screenshot shows the Microsoft Sentinel KQL query interface. The query is:


```

1 Event
2 | where TimeGenerated > ago(24h)
3 | where EventLog == "Microsoft-Windows-Sysmon/Operational"
4 | where RenderedDescription contains "94.158.244.69"
5 | project TimeGenerated, Computer, EventID, RenderedDescription
6 | order by TimeGenerated desc
  
```

 The results table shows one entry:

TimeGenerated [UTC]	Computer	EventID	RenderedDescription
2026-06-23T03:32:45.9013065Z	Web01	1	Process Create: RuleName: - UtcTime: 2026-06-23 03:32:45.890 ProcessGuid: {634acd6d-fe5d-6a39-d701-000000000000} ProcessId: 2908 Image: C:\Windows\System32\nslookup.exe FileVersion: 10.0.26100.7019 (WinBuild.160101.0800) Description: nslookup.exe

 The interface also shows a search bar at the top, a 'Run' button, and a 'Show: 1000 results' dropdown. The bottom status bar indicates '0s 609ms' and 'Display time (UTC+00:00)'.

Fig 12 -- Sentinel Logs: IOC Hunt results -- 1 result for 94.158.244.69 -- nslookup.exe (C:\Windows\System32\nslookup.exe) captured, RenderedDescription showing full process details

Hunt 2 -- Sysmon Image Load: LummaC2 DLL Indicators (Event ID 7)

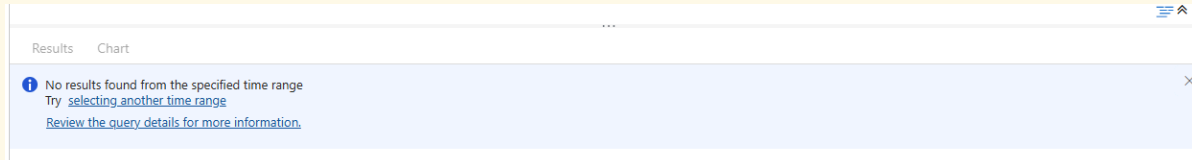
Hunts for Sysmon Event ID 7 (Image Loaded) where LummaC2-associated DLLs were loaded: iphlpapi.dll and winhttp.dll (per CISA Advisory Table 7).

```

Event
| where TimeGenerated > ago(24h)
| where EventLog == "Microsoft-Windows-Sysmon/Operational"
| where EventID == 7
| where RenderedDescription contains "iphlpapi.dll"
|   or RenderedDescription contains "winhttp.dll"
| project TimeGenerated, Computer, RenderedDescription
| order by TimeGenerated desc

```

No results -- as expected. The simulation does not load these DLLs. In a real LummaC2 infection, this query would surface DLL load events.

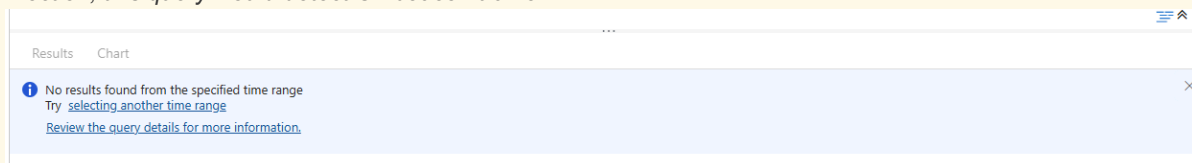


Hunt 3 -- Suspicious Network / Web Activity (Event ID 3)

Hunts for Sysmon Event ID 3 (Network Connection) originating from LummaC2.exe -- detecting C2 communication attempts.

```
Event
| where TimeGenerated > ago(24h)
| where EventLog == "Microsoft-Windows-Sysmon/Operational"
| where EventID == 3
| where RenderedDescription contains "LummaC2.exe"
| project TimeGenerated, Computer, RenderedDescription
| order by TimeGenerated desc
```

No results -- as expected. LummaC2.exe only launches calc.exe and makes no network connections. In a real infection, this query would detect C2 beacon traffic.



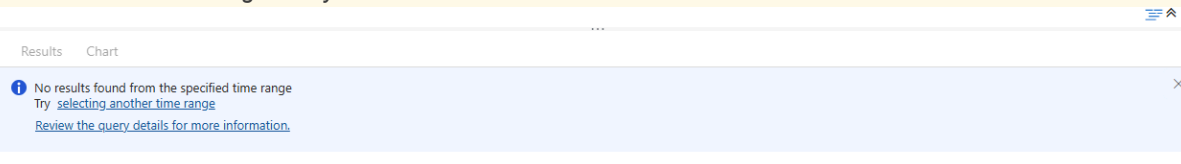
Hunt 4 -- DNS Query for Known LummaC2 Domains (Event ID 22)

Hunts for Sysmon Event ID 22 (DNS Query) against a subset of historically observed LummaC2 C2 domains from the CISA advisory.

```
let LummaDomains = dynamic([
    "blast-hubs.com",
    "blastikcn.com",
    "friendseforever.help",
    "shiningrstars.help",
    "generalmills.pro",
    "hoyoverse.blog"
]);
Event
| where TimeGenerated > ago(24h)
| where EventLog == "Microsoft-Windows-Sysmon/Operational"
| where EventID == 22
| where RenderedDescription has_any (LummaDomains)
| project TimeGenerated, Computer, RenderedDescription
```

```
order by TimeGenerated desc
```

No results -- as expected. No DNS queries were made to these C2 domains. In a real compromise, this query would surface beaconing activity.

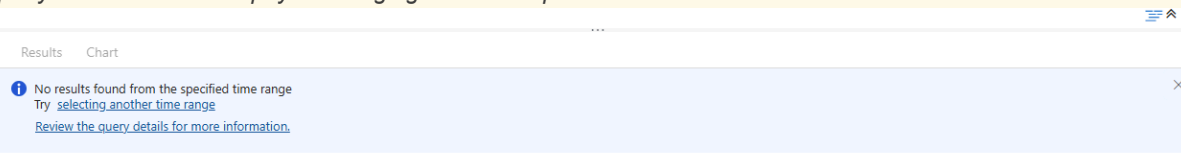


Hunt 5 -- Suspicious Download / Transfer Activity (Event ID 4688)

Hunts for Windows Security Event ID 4688 (New Process Created) where the command line includes common download tools used by LummaC2 for staged payload delivery.

```
SecurityEvent
| where TimeGenerated > ago(24h)
| where EventID == 4688
| where CommandLine has_any (
    "Invoke-WebRequest", "curl", "wget", "bitsadmin", "certutil"
)
| project TimeGenerated, Computer, Account, NewProcessName, CommandLine
| order by TimeGenerated desc
```

No results -- as expected. No download tools were used in the simulation. In a real LummaC2 infection, this query would catch initial payload staging via web requests.



10. Threat Hunt Summary

Hunt	Description	Sysmon Event	MITRE Technique	Result
1	LummaC2.exe Process Execution	Event ID 1	T1036 Masquerading	3 results -- CONFIRMED
IOC	nslookup 94.158.244.69	Event ID 1	T1046 Network Discovery	1 result -- CONFIRMED
2	DLL Image Load (iphlpapi/winhttp)	Event ID 7	T1574 DLL Side-Loading	No results (expected)
3	Network/C2 Activity	Event ID 3	T1071 Web Service C2	No results (expected)
4	DNS LummaC2 Domains	Event ID 22	T1071 DNS C2	No results (expected)
5	Download Tool Usage	Event ID 4688	T1105 Ingress Tool Transfer	No results (expected)

Hunt 1 and the IOC Investigation Hunt both returned results, confirming that the simulated LummaC2 telemetry was successfully ingested into Sentinel and was detectable via KQL. Hunts 2-5 returned no results

as expected since the simulation only generates process creation events -- not DLL loads, network connections, or DNS queries to C2 domains.

11. Skills Demonstrated

Hypothesis-Driven Threat Hunting	CISA Advisory-Based IOC Research	Sysmon Installation & Configuration
Custom DCR for Sysmon Logs	KQL Advanced Hunting Queries	Endpoint Telemetry Analysis (Event ID 1 confirmed; queries written for IDs 3, 7, 22)
IOC Investigation (nslookup)	MITRE ATT&CK Technique Mapping	PS2EXE Simulation Engineering
Microsoft Sentinel Log Analysis	Threat Intelligence Integration	Detection Engineering Validation

12. Conclusion

This lab demonstrated a complete hypothesis-driven threat hunting workflow aligned with the CISA cybersecurity advisory AA25-141b on LummaC2 malware. Starting from the hypothesis that LummaC2-like activity may have occurred in the environment, a safe simulation executable (LummaC2.exe) was created using PS2EXE to generate realistic process telemetry under a known IOC filename.

Sysmon v15.21 was installed on Web01 to provide deep endpoint visibility, and a custom DCR was configured to ingest Sysmon logs into Microsoft Sentinel. Five KQL hunting queries were written and executed against CISA advisory indicators -- Hunt 1 (LummaC2.exe process execution) returned 3 confirmed results, and the IOC investigation hunt (nslookup 94.158.244.69) returned 1 confirmed result. Hunts 2-5 returned no results as expected, since the simulation does not replicate the full LummaC2 execution chain.

In a real-world environment where LummaC2 was actually deployed, these same hunting queries would surface: DLL image loads (iphlpapi.dll, winhttp.dll), network connections to C2 infrastructure, DNS queries to known LummaC2 domains, and download tool usage for staged payload delivery -- providing a complete detection coverage framework aligned with the CISA advisory.

References

CISA Advisory AA25-141b -- <https://www.cisa.gov/news-events/cybersecurity-advisories/aa25-141b>
Microsoft Sysmon v15.21 -- <https://learn.microsoft.com/sysinternals/downloads/sysmon>
PS2EXE Module -- <https://github.com/MScholtes/PS2EXE>
LummaC2 any.run Analysis -- <https://any.run/malware-trends/lumma>
MITRE ATT&CK T1036 Masquerading -- <https://attack.mitre.org/techniques/T1036/>
Microsoft Sentinel KQL Reference -- <https://learn.microsoft.com/azure/data-explorer/kql-quick-reference>